

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 1 SET D

SUMMER SEMESTER, 2024-2025

DURATION: 30 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. Simplify the following boolean expression using cut and try method:

4 + 1
(CO2)
(PO1)

(i) $F_1(P, Q, R) = P\bar{Q}R + \bar{P}\bar{Q}\bar{R} + \bar{P}Q\bar{R} + P\bar{Q}\bar{R} + \bar{P}\bar{Q}R$

(ii) $F_2(P, Q) = (P \oplus Q) + \overline{(P \oplus Q)}$

Solution:**(i)**

$$\begin{aligned} F_1 &= P\bar{Q}R + \bar{P}\bar{Q}\bar{R} + \bar{P}Q\bar{R} + P\bar{Q}\bar{R} + \bar{P}\bar{Q}R \\ &= \bar{Q}(PR + \bar{P}\bar{R} + \bar{P}R + \bar{P}R) + \bar{P}Q\bar{R} \quad [\text{Distributive}] \\ &= \bar{Q}(P(R + \bar{R}) + \bar{P}(\bar{R} + R)) + \bar{P}Q\bar{R} \quad [\text{Distributive}] \\ &= \bar{Q}(P + \bar{P}) + \bar{P}Q\bar{R} \quad [\text{Complement}] \\ &= \bar{Q} + \bar{P}Q\bar{R} \quad [\text{Identity}] \\ &= \bar{Q} + \bar{P}\bar{R} \quad [\text{Absorption}] \end{aligned}$$

(ii)

$$F_2 = (P \oplus Q) + \overline{(P \oplus Q)} = T + T' = 1 \quad [A + A' \text{ form}]$$

Final Answer:

$$F_1 = \bar{Q} + \bar{P}\bar{R}, \quad F_2 = 1$$

Rubrics:

- 2 points for correct grouping/simplification steps for F_1 .
- 2 points for the final simplified form of F_1 .
- 1 point for identifying $F_2 = 1$.

2. Perform the following subtraction using the (r-1)'s complement method without converting the numbers to another base:

4
(CO1)
(PO1)

$$(105)_9 - (472)_9$$

Solution:

Given: $(0105)_9 - (0472)_9$

8's Complement Method (Diminished Radix Complement)

8's complement of $(0472)_9 = (8416)_9$

$$\begin{array}{r} 0 \ 1 \ 0 \ 5 \\ + \ 8 \ 4 \ 1 \ 6 \\ \hline 8 \ 5 \ 2 \ 2 \end{array}$$

Observation: No end carry occurs $\Rightarrow$ result is negative.

8's complement of $(8522)_9 = (0366)_9$

$$\Rightarrow \boxed{-(0366)_9}$$

Rubrics:

- 1 point for correct 8's complement of the subtrahend.
- 2 points for correct base-9 addition and end carry handling.
- 1 point for the final result.

3. Perform the following addition using BCD addition (you have to convert the numbers to BCD, perform the addition, and convert the result back to decimal):

5
(CO1)
(PO1)

$$(689)_{10} + (795)_{10}$$

Solution:

$$689_{10} = 0110 \ 1000 \ 1001_{BCD}$$

$$795_{10} = 0111 \ 1001 \ 0101_{BCD}$$

$$\begin{array}{r} 0 \ 1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \\ + \ 0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \\ \hline 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1^+ \ 1 \ 1 \ 1 \ 0 \\ + \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \\ \hline 1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \end{array}$$

Final Answer:

$$1 \ 0100 \ 1000 \ 0100_{BCD} = 1484_{10}$$

Rubrics:

- 2 points for correct BCD conversion.
- 2 points for correct BCD addition with corrections.
- 1 point for the final decimal result.

4. A number $(785)_b$ in an unknown base b is equivalent to the hexadecimal number $(284)_{16}$. Determine the base b .

3
(CO1)
(PO1)

Solution:

Given:

$$(785)_b = (284)_{16}$$

Step 1: Convert hexadecimal to decimal

$$(284)_{16} = 2 \cdot 16^2 + 8 \cdot 16^1 + 4 \cdot 16^0 = 512 + 128 + 4 = 644$$

Step 2: Convert $(785)_b$ to decimal and solve

$$7b^2 + 8b + 5 = 644 \Rightarrow 7b^2 + 8b - 639 = 0$$

Using the quadratic formula:

$$b = \frac{-8 \pm \sqrt{8^2 - 4 \cdot 7 \cdot (-639)}}{2 \cdot 7} = \frac{-8 \pm \sqrt{64 + 17892}}{14} = \frac{-8 \pm \sqrt{17956}}{14} = \frac{-8 \pm 134}{14}$$

Evaluating both roots:

$$b = \frac{-8 + 134}{14} = \frac{126}{14} = \boxed{9}$$

$$b = \frac{-8 - 134}{14} \quad (\text{Not valid})$$

Final Answer:

$$\boxed{b = 9}$$

Rubrics:

- 1 point for correctly converting the two numbers to a known base.
- 1 point for correctly expressing the given numbers into a valid equation.
- 1 point for correctly determining the root of the equation and neglecting the invalid root.

5. Pick the correct option for the following question:

1 × 3
(CO1,2)
(PO1)

- (A) Can $(0110101)_{GRAY}$ and $(0111101)_{GRAY}$ be two consecutive numbers in the Gray code sequence?

☐ Yes

☐ No

- (B) Does $(1110010)_2$ maintain odd parity?

☐ Yes

☐ No

- (C) Let $F(P, Q, R) = \sum M(1, 3, 4, 6)$. Is $\bar{F} = \sum m(1, 3, 4, 6)$ true?

☐ Yes

☐ No

Solution:

Answers:

(A) Yes

(B) No

(C) No

Reasoning:

(A) They have the same length and differ by exactly one bit.

(B) $(1110010)_2$ has four 1s, so odd parity is not maintained.

(C)

(D) Here F is 1, so the $\bar{F}$ needs to be 0 which is not in this case. Note that in the question, its **Sum of Maxterms** not the conventional **Product** of Maxterms.

Rubrics:

- 1 point for each correct option.